LESSON 10.5

Math is Boundless

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can create and describe geometric designs using repetition, patterns, and rhythm, and use a pattern unit to predict what comes next.

WHY THIS MATTERS Geometry is an integral part of design. Geometric patterns have been used to decorate homes and buildings for centuries — in art, architecture, and clothing. This lesson looks at three elements of geometric design: repetition, pattern, and rhythm.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
repetition *Spanish: repetición*	The same object, shape, or form repeated multiple times.	The same tile, over and over, all the way along the border.
pattern *Spanish: patrón*	A set of objects or shapes that are repeated and arranged in a regular order.	Circle, square, triangle, circle, square, triangle …
pattern unit *Spanish: unidad del patrón*	The smallest chunk of a pattern that repeats.	circle · square · triangle — the chunk that starts over.
rhythm *Spanish: ritmo*	A design element created by varying the placement, size, orientation, or order of the same objects or shapes.	The same circles, but sizes and spacing change as your eye travels.
predictability *Spanish: previsibilidad*	Being able to know what comes next because the design follows a rule.	Shape 25 must be a circle — the unit repeats every 3.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A border repeats: pink circle, purple square, blue triangle, pink circle, purple square, blue triangle, ...

1. The pattern unit is 3 shapes long: circle, square, triangle.
2. To predict shape number 12, I divide: 12 ÷ 3 = 4 with no remainder, so shape 12 ends a complete unit — it is the blue triangle.
3. The next three shapes after any complete unit will be a pink circle, a purple square, and a blue triangle.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

The pattern unit is still 3 shapes long.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The pattern unit is _____, so this is _____ rather than _____.
- I have seen a geometric pattern on _____ .
- Its pattern unit is _____ .
- A design with only repetition feels _____ .

Word bank repetition | pattern | pattern unit | rhythm | predictability

WATCH OUT!

A common mistake

Calling every repeated design a 'pattern.' One identical shape repeated is repetition

LESSON 10.5

Math is Boundless

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

In geometric design, what is repetition?

(A) A set of different shapes arranged randomly
(B) Shapes that vary in size and order of appearance
(C) Any drawing that uses color
(D) The same object or shape repeated multiple times

STEPS TO TRY
1. The word itself is a clue: something repeats.
2. How many different objects does repetition involve?
3. One object or shape, repeated multiple times.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A border shows: pink circle, purple square, blue triangle, pink circle, purple square, blue triangle. What are the next three shapes?

(A) Blue triangle, purple square, pink circle

(B) Pink circle, purple square, blue triangle

(C) Pink circle, blue triangle, purple square

(D) It is impossible to predict

STEPS TO TRY

1. Find the chunk that repeats.
2. The unit is circle, square, triangle.
3. After a triangle, the unit starts over.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

In the circle-square-triangle border, how many shapes long is the pattern unit?

(A) 1 shape

(B) 2 shapes

(C) 3 shapes

(D) 6 shapes

STEPS TO TRY

1. Where does the pattern start over?
2. List the shapes until you see the first shape again.
3. Circle, square, triangle — then it restarts.

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

4 **Answer in complete sentences.**

Where have you seen geometric patterns decorating a building, clothing, or artwork? Describe the design.

SENTENCE STARTER I have seen a geometric pattern on ___ .
Its pattern unit is ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Pick a card you called rhythm and your partner called repetition. What is the pattern unit, and what would have to change for it to stop being predictable?

SENTENCE STARTER The pattern unit is ___, so this is ___ rather than ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help